

Ștoc Iustin

CONTACT

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Github

EDUCATİON

Engineering Degree – Faculty of Automation and Computer Science, Technical University of Cluj-Napoca, 2022–2026

Currently pursuing: M.Sc. in Applied Informatics for Complex Systems Engineering (IAISC) – Technical University of Cluj-Napoca, 2026–Present

Mathematics-Informatics – "Emanuil Gojdu" National College, 2017–2021

AWARDS & PARTICIPATION

2026 – Paper accepted at ICECER 2026 (IEEE, Istanbul) – "An Open-Source Urban Integration Platform with Real-Time Visualization and Online SUMO–TomTom Traffic Calibration for Smart Cities"

2025 – Speaker, International Conference – Budapest, presentation on fractional-order control in anesthesia

PROFILE

Automation engineer and M.Sc. student in Applied Informatics for Complex Systems Engineering at UTCN. I am looking for a developer role, whether full-stack, backend, frontend or embedded, or a technical project management position.
I have delivered complete projects end to end. My bachelor's thesis is an open-source smart city platform for Cluj-Napoca (Node.js, React, PostgreSQL/PostGIS, SUMO), and it was extended into an IEEE paper accepted at ICECER 2026. I have also built full-stack applications in ASP.NET, Angular and Java, as well as embedded and IoT systems on STM32, Arduino and LoRa. As Project Manager on a smart city traffic system, I designed the adaptive traffic-light logic, defined the technical architecture and negotiated with suppliers.
I bring solid technical skills, fast learning and communication that connects engineering and business.

EXPERIENCE

→2025-2026
Project Manager – Smart City Traffic Systems / DTS
End-to-end project coordination for a smart city traffic system in Oradea. Designed the technical architecture across two contract lots, selected reliable solutions aligned with the beneficiary's requirements, negotiated contracts with external suppliers and ran the cost analysis for all equipment. Developed the adaptive traffic-light logic, with scenarios linked to LPR (License Plate Recognition) cameras for real-time traffic volume estimation, and integrated alternative-route display on Variable Message Signs (VMS). Proposed follow-up solutions for new projects, including reversible lanes in Florești and a smart city system in Marghita.

→ Summer 2024
Server / Rod 'N' Reel Resort
Server role in a high-volume hospitality environment as part of the Work and Travel USA program. Strengthened communication skills and adapted quickly to a new work culture in an American resort setting.

→ 2021–2022
SMT Operator / Plexus
PCB assembly and SMT line operation in electronics manufacturing. Offered a promotion to Technician based on adaptability and fast learning; left to focus on university studies in Cluj.

2023 – Participant, National Student Mathematics Competition "Traian Lalescu"

2021 – IC3-GS4 Certificate, Key Applications, Living Online, Computing Fundamentals

2019 – DPIT Academy – Fortech, Full-stack project for a tourist route application

LANGUAGES

English

French

PROJECTS

1) AirSniffer 360 — Distributed Air Quality Monitoring System
→ Hardware-focused contribution to a student research grant project building a distributed air quality monitoring system. Stack: Arduino Mega, LoRa SX1278 for long-range communication, GPS, FastAPI/MySQL backend, and a React/Leaflet dashboard. Also contributed to the grant application (feasibility, impact/dissemination, budget) and prepared the technical conference demo, including EU/ANCOM regulatory compliance context.

2) Open-Source Urban Integration Platform for Smart Cities(Bachelor's Thesis → IEEE Publication, ICECER 2026)
→ Designed and built an open-source urban integration platform for Cluj-Napoca as bachelor's thesis work, using Node.js/Express, React/Vite/Leaflet, PostgreSQL/PostGIS, and a SUMO + TraCI simulation bridge. A velocity-alignment technique reduced aggregate speed estimation error (MAPE) from 140% to 6%. Extended into a full paper — accepted for oral presentation at ICECER 2026 (IEEE, Istanbul) — adding real-time visualization and online SUMO-TomTom traffic calibration.

3) AI-Based Traffic Optimization
→Intelligent traffic control solution using YOLOv8 for real-time object detection in video streams. Implemented traffic light control logic based on detected congestion patterns. Built in Python and OpenCV, combining image processing with automated control strategies.

4) ProMed
→ Full-stack web platform built with ASP.NET, Angular, and SQL Server for managing appointments in public hospitals. Includes doctor-patient scheduling, secure authentication, and an admin panel. Demonstrates competence in software architecture, REST API development, and database integration.

SKİLLS

Programming & Software
- Java — 8/10
- C/C++ — 9/10
- C# — 9/10
- Python (image processing) — 8/10
- SQL — 9/10

Systems & Control Engineering
- Embedded systems and microcontrollers (STM32) — 8/10
- Design and analysis of control systems (MATLAB) — 9/10
- PLC programming (Siemens TIA Portal) — 7/10

Project & Delivery
- Project Management 9/10
- Vendor Negotiation 9/10